

Calculus II

Tangents and curve length

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Outline

- 1 Tangents to Curves
 - Tangents to Polar Curves

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 - Tangents to Polar Curves

- 2 Arc Length
 - Arc Length in Polar Coordinates

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Tangents

Let C be the curve $C : \begin{cases} x = f(t) \\ y = g(t) \end{cases}, t \in [a, b]$.

Definition

Suppose $f'(t)$ and $g'(t)$ are not simultaneously equal to 0.

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- We define the line passing through $(f(t), g(t))$ with direction vector equal to the tangent vector to be *tangent line* to C at t . In other words, the tangent line has equation

$$(x - f(t))g'(t) = (y - g(t))f'(t) \quad .$$

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Note. When $f'(t) = g'(t) = 0$, for curves C with additional properties, natural definition(s) of tangent(s) do exist but are beyond Calc II.

Example

Find the tangent to the curve

$$\gamma : \begin{cases} x = \cos t \\ y = \sin t \end{cases}, t \in [0, 2\pi) \text{ at } t = \frac{\pi}{4}, t = \frac{2\pi}{3}, t = \pi.$$

Recall C : $\begin{cases} x = f(t) \\ y = g(t) \end{cases}$, $t \in [a, b]$, tangent vector at t is $(f'(t), g'(t))$.

Observation

If $\frac{dx}{dt} \neq 0$, we have $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

Recall C : $\begin{cases} x = x(t) \\ y = y(t) \end{cases}$, $t \in [a, b]$, tangent vector at t is $(x'(t), y'(t))$.

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- Suppose we could eliminate the parameter t and write $y = F(x)$ for some function F near the point $(x, y) = (x(t), y(t))$.

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$$\begin{array}{lcl}
 y & = & F(x) \\
 \frac{dy}{dt} & = & \frac{d}{dt}(F(x)) \\
 & = & \frac{dF}{dx} \frac{dx}{dt} = \frac{dy}{dx} \frac{dx}{dt} \\
 \frac{dy}{dx} & = & \frac{\frac{dy}{dt}}{\frac{dx}{dt}}
 \end{array}
 \quad \left| \begin{array}{l} \text{apply } \frac{d}{dt} \\ \text{use chain rule} \\ \text{divide by } x'(t) \end{array} \right.$$

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If $\frac{dx}{dt} \neq 0$, we have $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

Example

A curve C is defined by $x = t^2$, $y = t^3 - 3t$.

- 1 Show C traverses $(x, y) = (3, 0)$ for two values of t ; find the tangent slopes for both of these values.
- 2 Find the points on C where the tangents are horizontal or vertical.
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- Plug in $t = \pm\sqrt{3}$: $\frac{dy}{dx}|_{t=\pm\sqrt{3}} = \frac{3(\pm\sqrt{3})^2 - 3}{2(\pm\sqrt{3})} = \pm \frac{6}{2\sqrt{3}} = \pm\sqrt{3}$

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- Therefore the tangents at $(3, 0)$ have slopes $\pm\sqrt{3}$.

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A curve C is defined by $x = t^2, y = t^3 - 3t$.

- ② Find the points on C where the tangents are horizontal or vertical.

Horizontal tangent:

$$\frac{dy}{dt} = 0$$

Vertical tangent:

$$\frac{dx}{dt} = 0$$

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The points are $(1, 2)$ and $(1, -2)$.

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The point is $(0, 0)$.

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Find the second derivative:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

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A curve C is defined by $x = t^2, y = t^3 - 3t$.

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Consider the cycloid $x = r(\theta - \sin \theta)$, $y = r(1 - \cos \theta)$.

- 1 At what points is the tangent horizontal?
- 2 At what points is the tangent vertical?

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- Therefore there is a horizontal tangent when $\theta = (2n + 1)\pi$.

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- Therefore there is a vertical tangent when $\theta = 2n\pi$.

Tangents to Polar Curves

To find the tangent line to a polar curve $r = f(\theta)$, regard θ as a parameter and write the parametric equations as

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The Arc Length Formula

Let $\gamma : \begin{cases} x &= x(t) \\ y &= y(t) \end{cases}, t \in [a, b]$.

Definition

Suppose $x'(t)$ and $y'(t)$ (exist and) are continuous on $[a, b]$. Then the length of the curve γ is defined as

$$L(\gamma) = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

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Arc length of graph of a function

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What is the length of the graph of the curve given by the graph of $y = f(x)$?

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Find the length of the arc of $y^2 = x^3$ between $(1, 1)$ and $(4, 8)$.

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$$\begin{aligned} L &= \int_1^4 \sqrt{1 + (y')^2} dx \\ &= \int_1^4 \sqrt{1 + \frac{9}{4}x} dx = \int_{13/4} \frac{4}{9} \sqrt{u} du \end{aligned}$$

Example

Find the length of the arc of $y^2 = x^3$ between $(1, 1)$ and $(4, 8)$.

- For the top half of the curve we have:
- $y = x^{3/2}$ and $y' = \frac{3}{2}x^{1/2}$.
- $u = 1 + \frac{9}{4}x$ and $du = \frac{9}{4}dx$.
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- When $x = 4$, $u = \quad$.

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Example

Find the length of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

$$L = \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Example

Find the length of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

$$L = \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{x=0}^{x=1} \sqrt{1 + \frac{dy}{dx}^2} dx = \int_{x=0}^{x=1} \sqrt{1 + ?} dx$$

Example

Find the length of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

$$L = \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{x=0}^{x=1} \sqrt{1 + \frac{dy}{dx} = 2x} \, dx$$

Example

Find the length of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

$$L = \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{x=0}^{x=1} \sqrt{1 + 4x^2} dx$$

$\frac{dy}{dx} = 2x$

Example

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$$L = \int_{x=0}^{x=1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{x=0}^{x=1} \sqrt{1 + 4x^2} dx \quad \left| \quad \frac{dy}{dx} = 2x \right. \quad \left| \quad \text{Set } x = \textcolor{red}{?} \tan \theta \right.$$

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 &= \frac{1}{2} \cdot \left[\frac{1}{2} (\sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|) \right]_{\theta=0}^{\theta=\alpha}
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 &= \frac{1}{4} (\sec \alpha \tan \alpha + \ln |\sec \alpha + \tan \alpha|) \\
 &= \frac{1}{4} \left(? \cdot ? + \ln |? + ?| \right)
 \end{aligned}$$

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 &= \frac{1}{4} (2 \cdot \sqrt{5} + \ln |\sqrt{5} + 2|)
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Example

Find the length of the curve γ .

$$\gamma : \begin{cases} x(t) = \sqrt{t} - 2t \\ y(t) = \frac{8}{3}t^{\frac{3}{4}} \end{cases}, t \in [1, 4] \quad .$$

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We have that $x'(t) = ?$ and $y'(t) = ?$

$$L(\gamma) = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt = \int_1^4 \sqrt{\left(? \right)^2 + \left(? \right)^2} dt$$

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$$\begin{aligned} L(\gamma) &= \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt = \int_1^4 \sqrt{\left(\frac{1}{2\sqrt{t}} - 2\right)^2 + \left(2t^{-\frac{1}{4}}\right)^2} dt \\ &= \int_1^4 \sqrt{\text{?} + \text{?}} dt \end{aligned}$$

Example

Find the length of the curve γ .

$$\gamma : \begin{cases} x(t) = \sqrt{t} - 2t \\ y(t) = \frac{8}{3}t^{\frac{3}{4}} \end{cases}, t \in [1, 4] .$$

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Find the length of the arc of $y = \frac{1}{6}e^{3x} + \frac{1}{6}e^{-3x}$ from $x = 0$ to $x = 1$.

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Find the length of the arc of $y = \frac{1}{6}e^{3x} + \frac{1}{6}e^{-3x}$ from $x = 0$ to $x = 1$.

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Find the length of one arch of the cycloid

$$x = r(\theta - \sin \theta), \quad y = r(1 - \cos \theta).$$

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Arc Length

To find the arc length of a polar curve $r = f(\theta)$, $a \leq \theta \leq b$, regard θ as a parameter.

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Find the length of the cardioid $r = 1 + \sin \theta$. The full length is given by $0 \leq \theta \leq 2\pi$.

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 &= -2 \left(- \right) + 2 \left(- \right) - 2 \left(- \right)
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 &= -2(0 - 2) + 2(2 - 0) - 2(0 - 2)
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 &= \int_0^{\pi/2} \frac{2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta + \int_{\pi/2}^{3\pi/2} \frac{-2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta + \int_{3\pi/2}^{2\pi} \frac{2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta \\
 &= \left[-2\sqrt{2 - 2\sin \theta}\right]_0^{\pi/2} + \left[2\sqrt{2 - 2\sin \theta}\right]_{\pi/2}^{3\pi/2} + \left[-2\sqrt{2 - 2\sin \theta}\right]_{3\pi/2}^{2\pi} \\
 &= -2(0 - \sqrt{2}) + 2(2 - 0) - 2(\sqrt{2} - 2)
 \end{aligned}$$

Example

Find the length of the cardioid $r = 1 + \sin \theta$. The full length is given by $0 \leq \theta \leq 2\pi$.

$$\begin{aligned}
 L &= \int_0^{2\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(1 + \sin \theta)^2 + \cos^2 \theta} d\theta \\
 &= \int_0^{2\pi} \frac{\sqrt{2 + 2\sin \theta} \sqrt{2 - 2\sin \theta}}{\sqrt{2 - 2\sin \theta}} d\theta = \int_0^{2\pi} \frac{\sqrt{4 - 4\sin^2 \theta}}{\sqrt{2 - 2\sin \theta}} d\theta \\
 &= \int_0^{2\pi} \frac{\sqrt{4\cos^2 \theta}}{\sqrt{2 - 2\sin \theta}} d\theta = \int_0^{2\pi} \frac{2|\cos \theta|}{\sqrt{2 - 2\sin \theta}} d\theta \\
 &= \int_0^{\pi/2} \frac{2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta + \int_{\pi/2}^{3\pi/2} \frac{-2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta + \int_{3\pi/2}^{2\pi} \frac{2\cos \theta}{\sqrt{2 - 2\sin \theta}} d\theta \\
 &= \left[-2\sqrt{2 - 2\sin \theta}\right]_0^{\pi/2} + \left[2\sqrt{2 - 2\sin \theta}\right]_{\pi/2}^{3\pi/2} + \left[-2\sqrt{2 - 2\sin \theta}\right]_{3\pi/2}^{2\pi} \\
 &= -2(0 - \sqrt{2}) + 2(2 - 0) - 2(\sqrt{2} - 2) = 8
 \end{aligned}$$